Validation Timeline — pqc_crypto_module v0.1.0

> **Disclaimer**: Prepared for FIPS 140-3 evaluation, not currently validated.

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1. Phase Overview

Phase	Duration (Est.)	Description
1. Lab Selection	2 weeks	Evaluate labs, send contact packages, receive proposals
2. Lab Onboarding	2-4 weeks	Contract signing, NDA, scope definition, kickoff meeting
3. Pre-Testing	4-8 weeks	ACVP harness build, CAVP submissions, documentation review with lab
4. Formal Testing & Iterations	2-6 months	Lab testing, findings resolution, re-testing cycles
5. CMVP Queue & Review	6-12 months	CMVP processes the validation report; may request clarifications
Total Estimated	**12-24 months**	From lab selection to FIPS 140-3 certificate issuance

2. Detailed Phase Breakdown

Phase 1: Lab Selection (Weeks 1-2)

Milestone	Target	Owner
Send contact packages to 2-3 labs	Week 1	[TBD]
Receive lab proposals and cost estimates	Week 2	[TBD]

Milestone	Target	Owner
Select lab and notify	Week 2	[TBD]

Deliverables: Signed letter of intent or contract initiation.

Phase 2: Lab Onboarding (Weeks 3-6)

Milestone	Target	Owner
Execute NDA and services agreement	Week 3	[TBD]
Kickoff meeting: scope, timeline, deliverables	Week 4	[TBD]
Receive lab-specific documentation requirements	Week 5	[TBD]
Submit initial documentation package for review	Week 6	[TBD]

Deliverables: Executed contract, agreed scope, documentation submission.

Phase 3: Pre-Testing (Weeks 7-14)

Milestone	Target	Owner
Build ACVP test harness	Weeks 7-8	Engineering
Integrate official NIST SHA-3 test vectors	Week 8	Engineering
Submit SHA-3 for CAVP certificate	Week 9	Lab
Address lab documentation feedback (round 1)	Weeks 10-12	Engineering
Integrate ML-DSA/ML-KEM ACVP vectors (if available)	Weeks 12-14	Engineering
Submit PQC algorithms for CAVP certificates	Week 14	Lab

Deliverables: ACVP harness, CAVP submissions, revised documentation.

Phase 4: Formal Testing & Iterations (Months 4-9)

Milestone	Target	Owner
Lab begins formal FIPS 140-3 testing	Month 4	Lab
Receive initial test findings	Month 5	Lab
Remediate findings (code or documentation)	Months 5-7	Engineering
Lab re-testing after remediation	Months 7-8	Lab
Final test report prepared by lab	Month 9	Lab

Deliverables: Passing test results, final validation report.

Phase 5: CMVP Queue & Review (Months 10-21)

Milestone	Target	Owner
Lab submits validation report to CMVP	Month 10	Lab
CMVP assigns reviewer	Months 10-12	CMVP
CMVP review and potential questions	Months 12-18	CMVP / Engineering
FIPS 140-3 certificate issued	Months 18-21	CMVP

Deliverables: FIPS 140-3 validation certificate.

3. Key Milestones

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Month 0        Lab selected
Month 1        Onboarding complete, documentation submitted
Month 2-3      ACVP harness ready, CAVP submissions initiated
Month 4-5      SHA-3 CAVP certificate obtained
Month 6-9      Formal testing and iterations
Month 9-10     Validation report submitted to CMVP
Month 18-21    Certificate issued (optimistic)
Month 24       Certificate issued (conservative)
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4. Risk Factors

Risk	Impact	Likelihood	Mitigation
NIST ACVP PQC support delayed	Blocks ML-DSA/ML-KEM CAVP certificates	Medium	Begin with SHA-3; proceed with PQC when available
ML-KEM-768 Rust crate not available	Blocks ML-KEM CAVP testing	Medium	Monitor ecosystem; consider C FFI wrapper as fallback
Lab findings require architectural changes	Adds 2-4 months	Low	Module architecture is well-aligned with FIPS requirements
CMVP queue backlog exceeds 12 months	Extends total timeline	Medium	Submit early; no mitigation for queue delays
Documentation revision cycles	Each cycle adds 2-4 weeks	Medium	Produce high-quality docs upfront (current state is strong)
Budget constraints	Delays entire timeline	Medium	Obtain budget approval before Phase 1

Risk	Impact	Likelihood	Mitigation
delay lab engagement			
Rust-specific lab unfamiliarity	Adds onboarding time	Low-Medium	Provide clear build instructions and reproducible environment

5. Cost Considerations

Typical FIPS 140-3 Level 1 software module validation costs:

Item	Estimated Range (USD)
Lab testing fees	\$50,000 - \$150,000
CAVP algorithm testing	\$10,000 - \$30,000 (per algorithm family)
Documentation review iterations	Included in lab fees (typically)
CMVP submission fee	Included in lab fees (typically)
Internal engineering effort	3-6 person-months
Total estimated	**\$80,000 - \$250,000**

Note: Costs vary significantly by lab, module complexity, and number of iteration cycles. PQC algorithms may carry a premium due to novelty. Request detailed quotes during Phase 1.

6. Success Criteria

- [ ] CAVP certificates obtained for ML-DSA-65, ML-KEM-768, SHA3-256
- [ ] Lab testing completed with all findings resolved
- [ ] Validation report accepted by CMVP
- [ ] FIPS 140-3 Level 1 certificate issued for `pqc_crypto_module v0.1.0`